

EF5342 Financial Systems, Markets and Instruments

Week 2 — Interest Rates

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Interest Rate Basics: Nominal vs. Real

Nominal interest rate (i) The rate actually observed in financial markets (bonds or loans)

Real risk-free rate (RFR) The true increase in purchasing power earned on a risk-free investment

Fisher equation: nominal rates contain compensation for expected inflation —
 $i \approx RFR + E(IP)$, exactly $1 + i = (1 + RFR)(1 + E(IP))$.

Example: 10-year Treasury yield 5%, expected inflation 3% $\Rightarrow$ $RFR \approx 5\% - 3\% = 2\%$ (exact: $1.05/1.03 - 1 \approx 1.94\%$). Your money grows 5% in dollar terms, but inflation erodes purchasing power by 3%, so real wealth rises only $\sim 2\%$.

In valuation (DCF, CAPM) use the **nominal** risk-free rate with nominal cash flows; with real (inflation-adjusted) cash flows, the **real** rate is the consistent choice. Nominal = what you see and get paid; real = what matters for long-term wealth growth after inflation.

Nominal Interest Rate and Inflation (US, 1986–2025)

Metric	Mean	Max	Min
10-Year Treasury yield (nominal risk-free)	5.1%	10.6% (1986)	0.5% (2020)
CPI inflation (YoY)	2.8%	8.0% (2022)	−0.4% (2009)
Real risk-free rate (nominal − inflation)	2.3%	7.0% (1986)	−5.5% (2022)

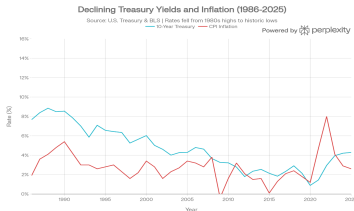
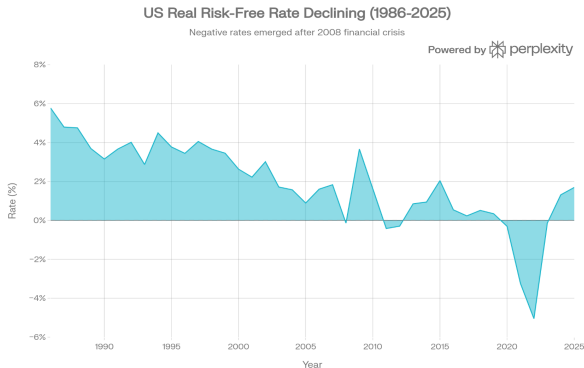


Table and chart show the 1986–2025 sample; as of 8 September 2026 the 10-year Treasury yield stands at 4.79%.

The Real Risk-Free Rate

The negative-rate era has ended: as of September 2026, no major central bank maintains a negative policy rate (ECB 1.50%, BoJ 0.75%, SNB 0.00%). The 2020–2023 episode was historically severe — pandemic-era monetary stimulus, supply-chain disruptions, and fiscal spending pushed inflation well above Treasury yields.



Loanable Funds Theory

Explains interest-rate *levels* and *movements*:

- The level of interest rates results from the **supply of and demand for loanable funds** — the interest rate is the *price* of loanable funds
- Market participants — consumers, businesses, governments, foreigners — are categorized as net suppliers or net demanders of funds
- The **equilibrium rate** i^* is where quantity supplied equals quantity demanded (Q^*)

Supply slopes upward, demand slopes downward; their intersection gives (i^*, Q^*) .

Supply and Demand of Loanable Funds

Supply — mainly savings (funds not consumed today):

- Household savings
- Business retained earnings
- Government budget surplus (public saving)
- Foreign capital inflows

Upward-sloping: higher real rates $\Rightarrow$ greater incentive to save (postpone consumption) $\Rightarrow$ more funds supplied.

Demand — from borrowers:

- Firms borrowing for physical capital (machinery, factories, R&D) — most important
- Households for big-ticket items (homes, cars)
- Government financing budget deficits

Downward-sloping: higher real rates $\Rightarrow$ borrowing costlier $\Rightarrow$ fewer profitable projects $\Rightarrow$ less funds demanded.

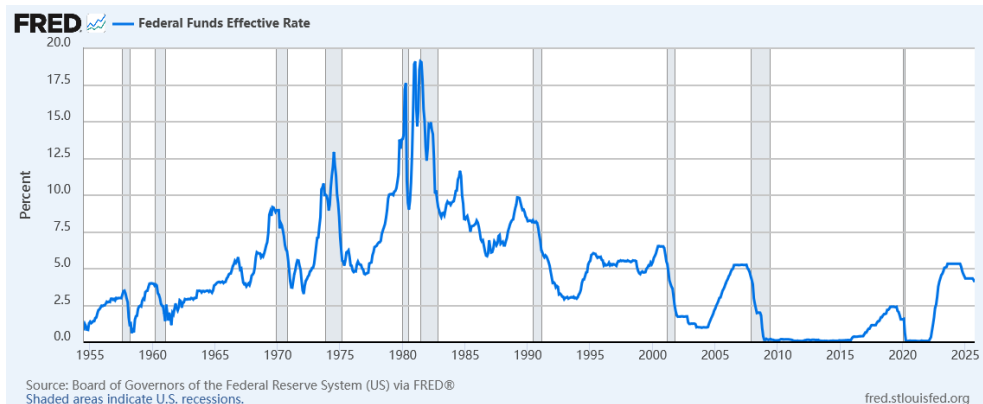
Shifts in the Curves Change Equilibrium Rates

Increased supply of loanable funds Supply shifts right ($SS \rightarrow SS^*$) — equilibrium moves $E \rightarrow E^*$: quantity rises ($Q^* \rightarrow Q^{**}$) and the interest rate *falls* ($i^* \rightarrow i^{**}$)

Increased demand for loanable funds Demand shifts right ($DD \rightarrow DD^*$) — equilibrium quantity rises ($Q^* \rightarrow Q^{**}$) and the interest rate *rises* ($i^* \rightarrow i^{**}$)

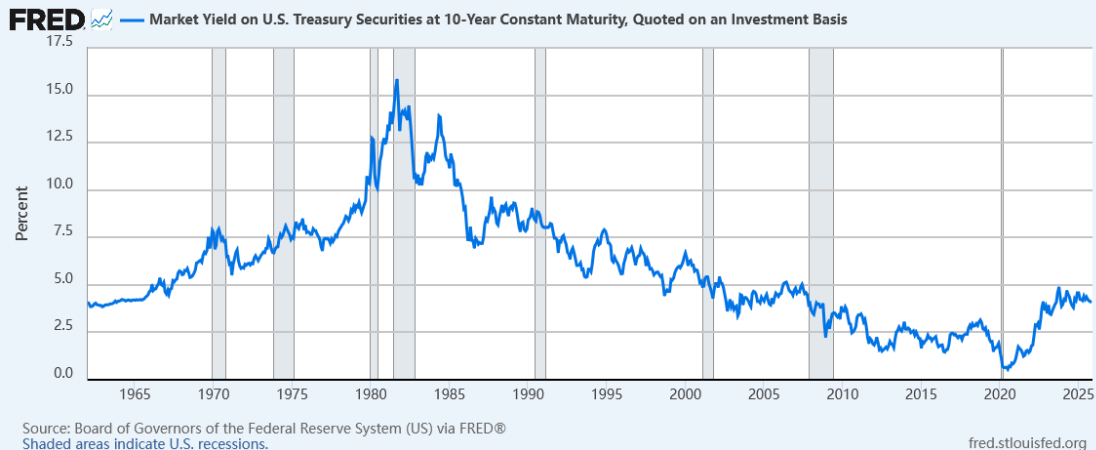
Key Interest Rates Over Time (US)

The **federal funds rate** is the rate at which depository institutions lend reserve balances to other banks overnight — the interest rate set by the Fed.



Source: Federal Reserve Bank of St. Louis

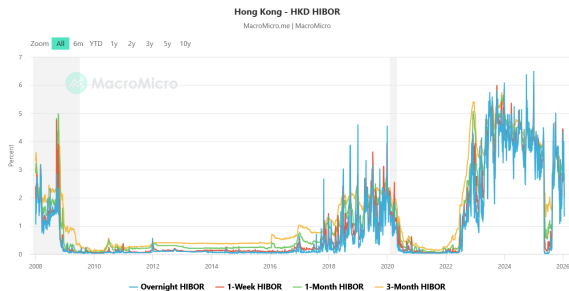
Key Interest Rates Over Time (US), cont.



Source: Federal Reserve Bank of St. Louis

Key Interest Rates in Hong Kong

The **HIBOR** (Hong Kong Interbank Offered Rate) is the main benchmark rate at which banks offer to lend HK dollar funds to other banks in the Hong Kong interbank market.



HKAB HKD Interest Settlement Rates (HKD HIBOR)

Specifications

Item	Specifications
Benchmark Owner	The Hong Kong Association of Banks (HKAB)
Benchmark Administrator	Treasury Markets Association
Calculation Agent	Hong Kong Interbank Clearing Limited (HKICL) Helpdesk: (852) 2533 1668
Definition	The estimated offer rates at which deposits in HKD for the Contract Period are being quoted to prime banks in the Hong Kong interbank market at 11:00a.m. on the relevant business day
Tenors	O/N, 1W, 2W, 1M, 2M, 3M, 6M and 12M
Contributing Banks	12-20 banks appointed by the HKAB and selected by the Hong Kong Monetary Authority (HKMA) based on banks' activeness in the different categories
Calculation Method	The benchmark rate is determined by averaging the middle quotes after excluding the highest three quotes and lowest three quotes received from the Contributing Banks.
Number of Decimal Places	5 decimal places
Contribution Window	Cut off at 11:10 a.m.

What Shifts the Curves? Supply-Side Factors

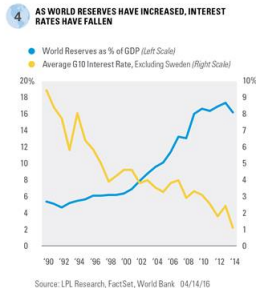
Increase in	Supply	Why / Result
Wealth & income	↑	More funds available for investment — <i>lower</i> interest rate
Risk	↓	Riskier securities are less attractive to fund suppliers — <i>higher</i> rate
Near-term spending needs	↓	Fewer funds available to invest — <i>higher</i> rate
Monetary expansion	↑	Central-bank money creation directly increases lendable funds — <i>lower</i> rate

What Shifts the Curves? Factors Moving Both Sides

Increase in	Supply	Demand	Why / Result
Economic growth	↑	↑	Wealth and income rise (more supply); more projects profitable (more demand) — effect on rate <i>indeterminate</i>
Utility from assets	↓	↑	Suppliers less willing to save; demanders more willing to borrow — <i>higher</i> rate
Restrictive covenants	↑	↓	Borrowers demand less; lenders more willing to supply — <i>lower</i> rate
Tax rate	↓	↑	Taxes reduce after-tax returns to savers; interest deductibility raises borrowing demand — <i>higher</i> rate

Case: The Global Savings Glut

“Whether it was a glut of excess intended saving, or a shortfall of investment intentions, the result was the same: a fall in global real long-term interest rates and their associated capitalization rates. Asset prices, particularly house prices, in nearly two dozen countries accordingly moved dramatically higher.” — Alan Greenspan, 2010



Determinants of Rates for Individual Securities

$$i_j^* = f(IP, RFR, DRP_j, LRP_j, SCP_j, MP_j)$$

where i_j^* = equilibrium nominal interest rate for security j .

Inflation premium (IP) $IP = \frac{CPI_{t+1} - CPI_t}{CPI_t} \times 100$; CPI = price level of a basket of goods and services purchased by the typical urban consumer

Real risk-free rate (RFR) $RFR = i - E(IP)$; measures society's relative time preference for consuming today rather than tomorrow

Default Risk Premium (DRP)

Compensation for the risk that the issuer fails to pay:

$$DRP_j = i_{jt} - i_{Tt}$$

where i_{jt} = interest rate on security j at time t and i_{Tt} = rate on a similar-maturity U.S. Treasury security at time t .



Remaining Security-Specific Premiums

Liquidity risk premium (LRP) Extra required yield to offset the risk of not being able to sell the asset in a timely fashion at fair value

Special feature premium (SCP) Taxability (municipal bonds exempt from federal/state taxes); convertibility (convertible bonds have lower yields); callability (callable bonds require higher yields)

Maturity premium (MP) Difference between yields on long- and short-term securities of the same characteristics

Inflation and the real risk-free rate are the **same for all securities**; DRP, LRP, SCP, and MP are **security-specific**.

Term Structure of Interest Rates (Yield Curve)

The **term structure of interest rates** (also called the **yield curve**) describes the relationship between interest rates (yields) and different maturities for bonds of similar credit quality (typically government bonds). Typical shapes, plotting yield to maturity against time to maturity:

- (a) Upward sloping yields rise with maturity (“normal”)
- (b) Inverted / downward sloping yields fall with maturity
- (c) Flat yields roughly equal across maturities

Three Theories of the Term Structure

Several major theories explain why the yield curve takes different shapes — upward-sloping (normal), flat, downward-sloping (inverted), or humped — and what drives long-term vs. short-term rates:

Unbiased expectations theory (UET) Long-term rates are purely the geometric average of current and expected future short-term rates

Liquidity premium theory Builds on pure expectations but adds a positive liquidity (term) premium that increases with maturity

Market segmentation theory Markets for different maturities are completely separate; investors and borrowers have strict maturity preferences and do not substitute across maturities

Unbiased Expectations Theory (UET)

Long-term rates are geometric averages of current and expected future short-term rates:

$${}_1R_N = [(1 + {}_1R_1)(1 + E({}_2r_1)) \cdots (1 + E({}_Nr_1))]^{1/N} - 1$$

with ${}_1R_N$ = annualized rate on an N -year bond, ${}_1R_1$ = actual one-year rate today, and $E({}_ir_1)$ = expected one-year rate from year $i - 1$ to i .

UET is a good benchmark: in equilibrium, the return from holding an N -year bond to maturity should equal the expected return from investing in N successive one-year bonds.

Constructing a Yield Curve with UET: Example

Suppose the current one-year rate and expected one-year rates are ${}_1R_1 = 1.94\%$, $E({}_2r_1) = 3\%$, $E({}_3r_1) = 3.74\%$, $E({}_4r_1) = 4.10\%$. Under UET, today's one-, two-, three-, and four-year Treasury rates are ${}_1R_1 = 1.94\%$ and:

$${}_1R_2 = [(1 + 0.0194)(1 + 0.03)]^{1/2} - 1 = 2.47\%$$

$${}_1R_4 = [(1 + 0.0194)(1 + 0.03)(1 + 0.0374)(1 + 0.041)]^{1/4} - 1 = 3.19\%$$

$${}_1R_3 = [(1 + 0.0194)(1 + 0.03)(1 + 0.0374)]^{1/3} - 1 = 2.89\%$$

Liquidity Premium Theory

Long-term rates are geometric averages of current and expected future short rates *plus a liquidity premium that increases with maturity*:

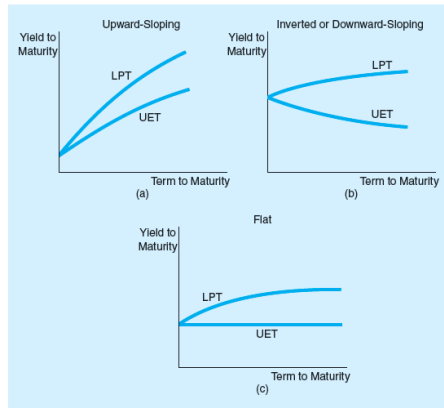
$${}_1R_N = [(1 + {}_1R_1)(1 + E({}_2r_1)) \cdots (1 + E({}_Nr_1)) + L_N]^{1/N} - 1$$

where L_t = liquidity premium for period t , with $L_2 < L_3 < \cdots < L_N$.

The theory implicitly assumes investors prefer short-term securities — more liquid, and prices less sensitive to interest-rate changes.

UET vs. Liquidity Premium Theory

Figure 2-9 Yield Curve under the Unbiased Expectations Theory (UET) versus the Liquidity Premium Theory (LPT)



Market Segmentation Theory

Investors and financial institutions have **specific maturity preferences**:

- Money market funds prefer short-term Treasury bonds
- Insurance companies and pension funds prefer longer-maturity bonds

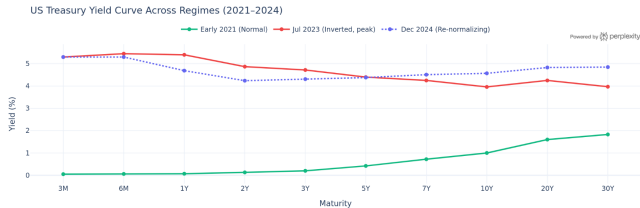
Yields in each maturity segment are determined *independently* by supply and demand within that segment; investors and borrowers deviate from their preferred segment only when adequately compensated. Under this theory, the long-term rate is **unrelated** to expected future short-term rates.

Comparison Between Theories

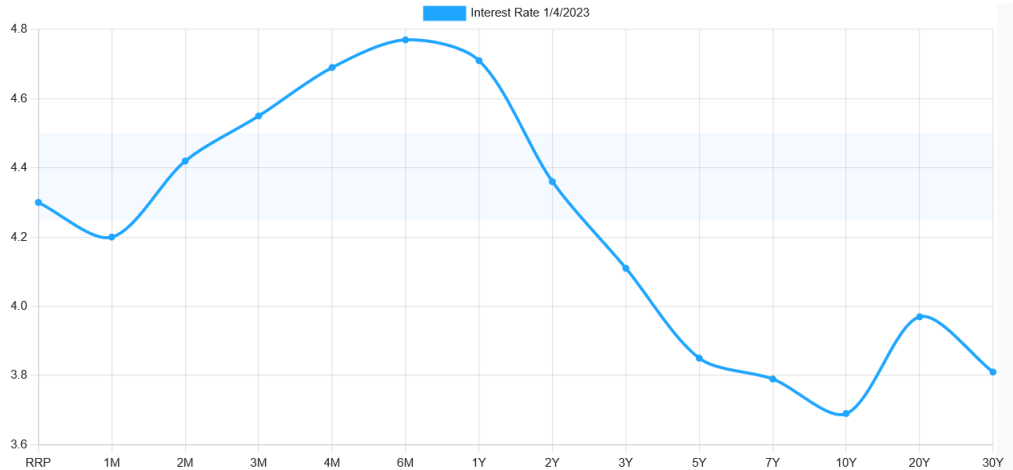
Theory		Key Driver	Risk/Term Premium?	Pre-	Typical Bias	Best explains. . .	Ex-
Unbiased expectations		Expectations of future short rates	None		Any shape (no bias)	Changes in curve shape over time	
Liquidity premium		Expectations + liquidity premium	Positive & increasing maturity	& in- with	Upward	Persistent upward slope historically	
Segmented markets		Supply & demand in maturity segments	Independent segment	per	Any shape	Supply/demand-driven anomalies	

Yield Curve Shapes: Stylized Analysis

Characteristic	Early 2021 (Normal)	Jul 2023 (Inverted)	Sep 2026 (Normal, flat)
3M yield	~0.05%	~5.30%	~3.87%
10Y yield	~1.00%	~3.96%	~4.79%
Slope (10Y–3M)	+ ~95bp	~−134bp	+ ~92bp
Signal	Expansion	Recession fear	Expansion (flat curve)



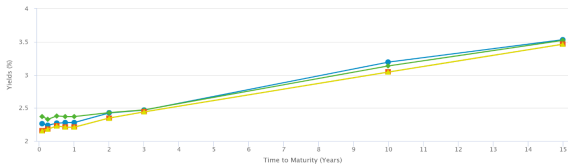
US Treasury Yield Curve



Government Bond Yield Curves in Other Markets

LCY Government Bond Yield Movements (HK)

Closing of 13-Jan-26



Source:AsianBondsOnline

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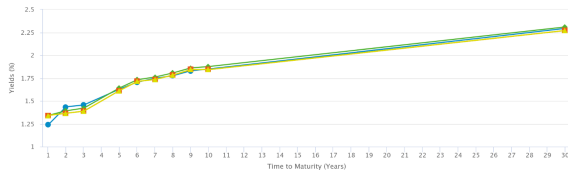
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LCY Government Bond Yield Movements (CN)

Closing of 13-Jan-26



Source:AsianBondsOnline

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Implied Forward Rates

A **forward rate** (f) is an expected rate on a short-term bond to be originated at some point in the future. UET states

$$(1 + {}_1R_N)^N = (1 + {}_1R_1)(1 + E({}_2r_1)) \cdots (1 + E({}_Nr_1))$$

Rearranging for adjacent maturities gives the one-year forward rate from year $N - 1$ to year N :

$${}_{N-1}f_1 = \frac{(1 + {}_1R_N)^N}{(1 + {}_1R_{N-1})^{N-1}} - 1$$

Implied Forward Rates: Example

Suppose (illustrative numbers, in the spirit of the low-rate 2010s) the spot one-, two-, three-, and four-year zero-coupon Treasury rates are ${}_1R_1 = 0.156\%$, ${}_1R_2 = 0.305\%$, ${}_1R_3 = 0.499\%$, ${}_1R_4 = 0.753\%$. Using UET, the one-year forward rates from year 1 to 2, 2 to 3, and 3 to 4 are:

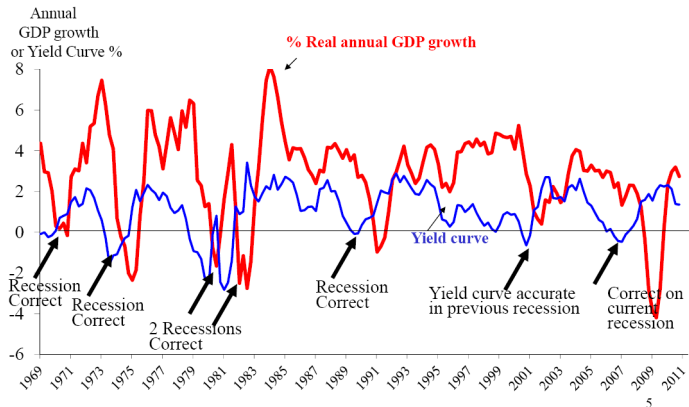
$${}_2f_1 = [(1 + 0.00305)^2 / (1 + 0.00156)] - 1 = 0.454\%$$

$${}_4f_1 = [(1 + 0.00753)^4 / (1 + 0.00499)^3] - 1 = 1.519\%$$

$${}_3f_1 = [(1 + 0.00499)^3 / (1 + 0.00305)^2] - 1 = 0.888\%$$

The Yield Curve Inverts Before Recessions

Spread: 5-year Treasury note minus 3-month Treasury bill.



Source: Campbell R. Harvey. Update of Harvey (1986, 1988, 1989, 1991).

Data through May 17, 2011

Source: duke.edu/~charvey

Mechanisms Behind the Predictability

Main mechanism — expectations of future Fed rate cuts due to economic weakness.

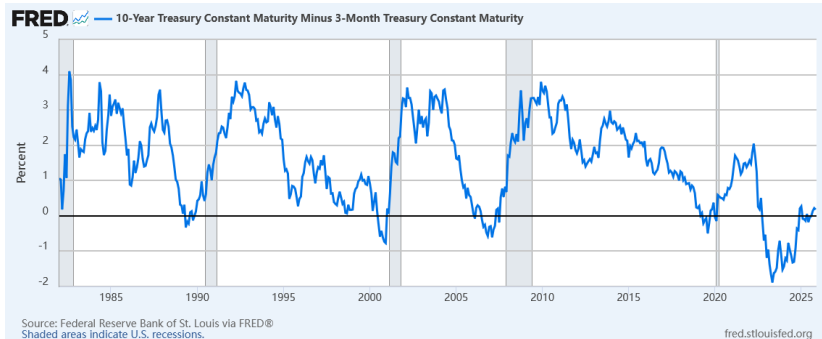
Bond markets price Treasuries off expected future short rates (heavily influenced by the federal funds rate) plus a term premium for risk and inflation uncertainty.

- Upward-sloping curve: investors expect short rates to rise (or stay stable) — typical of expansions
- Inverted curve: investors expect short rates to fall significantly — usually because they anticipate slowdown or recession

Secondary channel — tight policy and credit conditions. Inversions often follow aggressive Fed hikes: short yields jump while long yields rise less (or fall) on expected easing. Tightening slows the economy by raising borrowing costs and squeezing bank net interest margins (borrow short, lend long), leading banks to tighten lending standards; reduced credit amplifies weakness.

But the Signal Gave a False Alarm Recently...

The long 2022–2024 inversion was the longest on record and was not followed by an immediate recession, leading some to question its reliability — but the core logic still holds in most analyses. The curve has since returned to a normal shape: as of September 2026 it is positively sloped but unusually flat (10Y exceeds 2Y by only ~40 bp).



Source: FRED, T10Y3M

Time Value of Money: Lump Sums

A dollar received today is worth more than a dollar received in the future. **Simple interest:** interest is not reinvested. **Compound interest:** interest is reinvested (the standard assumption in finance).

Present value Discount a future payment at the current rate:

$$PV = FV_t \left[\frac{1}{1+r} \right]^t = FV_t \cdot PVIF_{r,t}$$

Future value Compound a present sum: $FV_t = PV(1+r)^t = PV \cdot FVIF_{r,t}$

where FV_t = future lump sum received in t periods, r = annual interest rate, t = years, and $PVIF_{r,t}$ ($FVIF_{r,t}$) = present (future) value interest factor of a lump sum.

Present Value of an Annuity

The present value of a finite series of equal cash flows received at the *end* of each equal interval (ordinary annuity):

$$PV = PMT \cdot PVIFA_{r,t}, \quad PVIFA_{r,t} = \sum_{n=1}^t \frac{1}{(1+r)^n}$$

If cash flows arrive on the *first* day of each period (annuity due), each payment earns one extra period of interest:

$$PV^{due} = PMT \cdot PVIFA_{r,t} \times (1+r)$$

Present Value of an Annuity: Example

A bond pays \$10,000 on the last day of every year for the next six years, in exchange for a fixed payment today. At an annual rate of 8%:

$$\begin{aligned} PV &= PMT \sum_{j=1}^t [1/(1+r)]^j = PMT \times \left[\frac{1 - (1+r)^{-t}}{r} \right] \\ &= \$10,000 \times \left[\frac{1 - (1 + 0.08)^{-6}}{0.08} \right] = \$46,228.80 \end{aligned}$$

If instead it pays \$10,000 on the last day of every *quarter* for six years ($r = 8\%/4 = 2\%$, $t = 6 \times 4 = 24$), the present value becomes:

$$\begin{aligned} PV &= \$10,000 \times \left[\frac{1 - (1 + 0.02)^{-24}}{0.02} \right] = \$10,000 * 18.913926 \\ &= \$189,139.26 \end{aligned}$$

Future Value of an Annuity

The future value of a finite series of equal cash flows received at the end of each interval:

$$FV_t = PMT \cdot FVIFA_{r,t}, \quad FVIFA_{r,t} = \sum_{n=1}^t (1+r)^{t-n} = \frac{(1+r)^t - 1}{r}$$

Future Value of an Annuity: Example

You want \$5 million when you retire in 40 years and believe you can earn 9% per year. How much must you invest each year, with the first investment at the end of the first year? (Ignore taxes.)

$$FV_t = PMT \sum_{j=0}^{t-1} (1+r)^j = PMT \times \left[\frac{(1+r)^t - 1}{r} \right]$$

$$\$5,000,000 = PMT \times \left[\frac{(1+0.09)^{40} - 1}{0.09} \right]$$

$$PMT = \$14,798$$

Key Takeaways

Determinants of interest rates Real risk-free rate + inflation expectations (Fisher effect: $i \approx RFR + E(IP)$); market equilibrium set by supply (savings) and demand (borrowing) of loanable funds; security-specific yields add premiums for default, liquidity, maturity, and special features

Term structure (yield curve) Relationship between yield and maturity for similar-risk bonds; theories — expectations (long rates = average of expected future short rates), liquidity premium (extra compensation for longer maturities), market segmentation (supply and demand within maturity segments set rates independently)

Key insight An inverted curve signals expected rate cuts and predicts recession

Grading Rubrics: Four Dimensions

Presentation (10 points)

Content and Ideas (4)	Poor (60%)	Satisfactory (75%)	Good (85%)	V. (95%)	Good
Effective Use of Resources (2)					
Creativity/Novelty (2)					
Presentation Skill (2)					

Report (20 points)

Content and Ideas (8)	Poor (60%)	Satisfactory (75%)	Good (85%)	V. (95%)	Good
Effective Use of Resources (4)					
Creativity/Novelty (4)					
Writing Clarity (4)					

Grading Rubrics Explained

1. **Content and ideas** The foundational substance and intellectual core — not just information, but a purposeful, insightful, well-structured argument or exploration
2. **Effective use of resources** How you select, integrate, and build on existing evidence — use authoritative, relevant sources (scholarly articles, reputable books, reliable data, primary sources); avoid weak sources such as unexplained web pages or popular forums
3. **Creativity / novelty** The unique perspective, innovative approach, or original contribution — ask your group: “What will make our project memorable and distinct from seven other groups doing a similar topic?”
4. **Writing clarity and presentation** The precision, polish, and professionalism of the communication — does the final product read as one unified voice? Seamless integration of different writers' sections is crucial